DeepLOB Reproduction and Cross-Market Transfer for Limit Order Book Forecasting

IEOR 242 Spring 2026 Final Project Report

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This repository version lists the submitting author only. Full team information and the presentation link can be added in the submission copy.

Abstract

This project studies whether a deep limit order book model can be reproduced on a standard benchmark and transferred to a materially different market setting. We reproduce DeepLOB on FI-2010 [4, 2] and adapt the same design to Optiver auction-book data [3], where only two book levels are visible. On FI-2010, the best short-horizon result reaches accuracy 0.8347 and MCC 0.6080 at 10 events, while the best long-horizon result reaches MCC 0.6943 at 100 events under an adaptive training profile. We also compare the deep model against shallow manual-factor baselines and analyze 104 engineered microstructure factors, finding that price and volume derivative features dominate the qualified factor set. On Optiver, we focus on the rolling-quantile label mode and compare zero-shot transfer, per-stock fine-tuning, and specific-stock out-of-sample training. Accuracy is modest, but it is not the right standalone criterion: fine-tuning improves Cohen's κ from 0.0618 to 0.0844 and MCC from 0.0626 to 0.0847, even when accuracy declines slightly. Cumulative normalized profit curves and a local LIME-style explanation provide complementary evidence. The main limitation is persistent nonstationarity: specific-stock out-of-sample training collapses to near-zero agreement, showing that transfer is constrained less by optimization than by distribution shift and label instability.

1 Project Description and Motivation

Limit order book (LOB) forecasting asks whether recent order-flow states contain enough information to predict short-horizon mid-price direction. We study a three-class version of this task: given a recent sequence of LOB states, predict whether the future mid-price will move down, remain stationary, or move up over a fixed number of events. The use-case is relevant to execution timing, market making, and short-horizon risk monitoring, though in this project predictions are treated as research diagnostics rather than deployable trading signals.

The problem is a strong fit for deep learning because the input has both spatial and temporal structure. DeepLOB [4] is attractive precisely because it models both: convolutional layers extract local cross-level patterns, an Inception-style block captures multiple local scales, and an LSTM aggregates temporal context. The project therefore asks a concrete question: can this architecture be reproduced on FI-2010 and then transferred to Optiver's shallower two-level auction book without redesigning the model family? That framing lets the report address the course requirements directly: a motivated use-case, a precise supervised learning problem, a justified methodology with advanced components, diagnostic analysis of results, and a safety/ethics discussion.

2 Data and Learning Problem

2.1 Dataset Overview

We work with two datasets that share the same prediction target but differ substantially in representation.

FI-2010 benchmark. FI-2010 [2] is a standard LOB benchmark with 40 normalized input features per event, representing 10 bid and ask price levels and their corresponding volumes. We use the standard three-class mid-price movement task at horizons of 10, 20, 30, 50, and 100 events ahead.

Optiver auction-book data. The Optiver workflow is built from processed per-stock files derived from the Optiver realized-volatility competition data [3]. Each event exposes only two order-book levels, yielding 8 reordered features:

$$\left\{ \begin{array}{l} \texttt{ask\_price1}, \texttt{ask\_size1}, \texttt{bid\_price1}, \texttt{bid\_size1}, \\ \texttt{ask\_price2}, \texttt{ask\_size2}, \texttt{bid\_price2}, \texttt{bid\_size2} \end{array} \right\}.$$

All reported Optiver experiments use the 5-event horizon and evaluate transfer on held-out stocks 42, 17, and 82.

Sample construction. Both datasets are converted into overlapping windows. Let $X_t \in \mathbb{R}^{100 \times d}$ denote the 100-event lookback window ending at event t, where $d = 40$ for FI-2010 and $d = 8$ for Optiver. The target label is $Y_t \in \{-1, 0, +1\}$, indicating down, stationary, or up movement over a future horizon of k events. The model learns a mapping

$$f_\theta : X_t \mapsto Y_t.$$

2.2 Label Construction and Preprocessing

For FI-2010, we follow the standard benchmark task. For Optiver, labels are constructed from future log mid-price returns using rolling-quantile thresholding: the class boundaries are the rolling historical 33rd and 67th percentiles of absolute k-step returns over a 20,000-sample window. This allows thresholds to adapt to stock-specific volatility scale and reduces the brittleness of fixed absolute thresholds across stocks.

The most important preprocessing design choice is causality. Optiver inputs use rolling normalization computed only from past observations, so each example is standardized without leakage from the future. All train, validation, and test splits are chronological rather than randomly shuffled. This is essential because random mixing would substantially overstate generalization in a nonstationary time-series setting.

2.3 Data Quality and Modeling Challenges

Three issues drive the difficulty of the project.

- **Class imbalance.** Short-horizon labels are often dominated by the stationary class, so accuracy alone can be misleading.
- **Representation mismatch.** FI-2010 exposes 10 levels, whereas Optiver exposes only two. This creates a genuine cross-representation transfer problem.

- **Temporal nonstationarity.** On Optiver, a model trained on one temporal segment of a stock can fail on a later segment of that same stock, even before cross-stock transfer is considered.

3 Methodology

3.1 Model Architecture

DeepLOB on FI-2010. Our FI-2010 model closely follows DeepLOB [4]. The input tensor has shape $(B, 1, T, 40)$ with $T = 100$. The architecture has three convolutional blocks followed by an Inception module, a one-layer LSTM, and a linear classifier. Concretely, the first block begins with a $(1, 2)$ stride-2 convolution to merge adjacent bid/ask entries, followed by two $(4, 1)$ convolutions; the second block repeats the same pattern; the third block uses a $(1, 10)$ convolution to collapse the remaining spatial dimension and then two additional $(4, 1)$ convolutions. The hidden activations are not uniform: blocks 1 and 3 use LeakyReLU(0.01), block 2 uses Tanh, and each stage is batch-normalized. The Inception module has three branches: $(1, 1) \to (3, 1)$, $(1, 1) \to (5, 1)$, and max-pooling followed by $(1, 1)$. These three branches are concatenated into a 192-dimensional feature sequence, which is passed to a one-layer LSTM with hidden size 64. A dropout layer regularizes the final representation before a fully connected layer outputs three logits, which are interpreted with Softmax at evaluation time.

DeepLOBLite on Optiver. For Optiver, we adapt the input interface to $(B, 1, T, 8)$ and preserve the overall CNN–Inception–LSTM design. The key architectural change is in the third convolutional block: after two stride-2 $(1, 2)$ reductions, the feature map width is only 2, so the FI-2010 $(1, 10)$ kernel is replaced by a $(1, 2)$ kernel to collapse the remaining spatial axis exactly. The remaining recurrent and classification layers are unchanged, including the 192-dimensional concatenated Inception output, the one-layer LSTM with hidden size 64, and the dropout-plus-linear head. This produces a lightweight transfer variant we refer to as DeepLOBLite.

3.2 Objective Function and Training

We train parameters θ by solving

$$\min_{\theta} \frac{1}{N} \sum_{i=1}^{N} L(f_\theta(X_i), Y_i) + R(\theta).$$

For FI-2010, L is multiclass cross-entropy and optimization uses Adam. The paper-style profile uses learning rate 10^{-3}, batch size 32, and validation-accuracy checkpointing for all horizons. The adaptive profile changes the monitoring rule and regularization by horizon, using stronger regularization and validation-loss monitoring at longer horizons where overfitting appears earlier.

For Optiver, we use focal loss with $\gamma = 1.5$, dropout of 0.30, weight decay of 10^{-4}, gradient clipping at 1.0, and macro-F1-based model selection. The implementation also includes an auxiliary log-return regression head to regularize the shared representation under the smaller two-level input.

These choices are problem-specific rather than generic defaults. Cross-entropy is appropriate for the FI benchmark because the task is standard multiclass classification and the goal is direct comparability to prior work. Adam is used because the gradients are noisy and the sequence model is moderately deep, making adaptive first-order optimization a pragmatic choice. On Optiver, focal loss is preferred because class imbalance and stationary-class dominance otherwise encourage majority-class collapse. Weight decay, dropout, early stopping, and gradient clipping are used to reduce high-variance behavior, while label smoothing is introduced only in the adaptive FI setup where long-horizon training is most prone to overfitting.

3.3 Advanced Components Beyond Class Material

The report includes two methodological elements that go beyond a direct lecture-style baseline: rolling-quantile label construction for Optiver, which adapts thresholds to stock-specific volatility, and horizon-aware training control for FI-2010, which changes checkpointing and regularization by horizon to address distinct short-horizon and long-horizon failure modes. These choices matter because the empirical bottleneck is not simply model capacity; it is calibration under skewed labels, changing return scales, and temporal drift.

3.4 Baselines and Comparisons

On FI-2010, we compare DeepLOB against ridge logistic regression and a two-layer MLP trained on 104 engineered microstructure features, following the spirit of earlier feature-based LOB work [1]. On Optiver, we compare three regimes: universal zero-shot transfer, per-stock fine-tuning, and specific-stock out-of-sample training. These comparisons let us separate the value of the architecture itself from the value of transfer calibration.

4 Experimental Setup

All experiments were run in a cluster-oriented workflow on PSC Bridges-2 using PyTorch. Training was launched through SLURM jobs, while notebooks were used only to summarize saved artifacts. Long GPU runs save checkpoints and metrics under `models/` and `results/`. The workflow uses PyTorch, NumPy, pandas, matplotlib, scikit-learn, SciPy, and statsmodels. Primary metrics are accuracy, Cohen's κ, and MCC; secondary diagnostics are training and validation loss curves, confusion matrices, cumulative normalized profit curves, and a local interpretability plot for a representative Optiver sample.

For this project, the most important evaluation choice is not to over-interpret accuracy. This is especially important on Optiver, where modest accuracy changes can hide meaningful changes in class balance, agreement beyond chance, and directional usefulness. To keep the narrative focused while still documenting the final choices, the appendix visualizes the retained FI configuration set (paper for 10/20/30 events, adaptive for 50/100 events) and the rolling3 Optiver diagnostics only.

5 Results and Discussion

5.1 FI-2010 Reproduction and Benchmark Analysis

Table 1 reports the FI-2010 headline results using the strongest profile for each horizon group.

Table 1: Primary FI-2010 results, using the paper profile for 10–30 events and the adaptive profile for 50–100 events.

Horizon	Selected profile	Accuracy	κ	MCC
10 events	Paper	0.8347	0.5907	0.6080
20 events	Paper	0.7506	0.4879	0.5060
30 events	Paper	0.7726	0.5980	0.6023
50 events	Adaptive	0.7940	0.6748	0.6751
100 events	Adaptive	0.7944	0.6920	0.6943

The most important FI-2010 result is not just that the model performs well, but that *different training rules are best at different horizons*. At short horizons, the paper-style profile is stronger because validation-accuracy checkpointing prevents the model from collapsing toward the stationary class. At long horizons, the adaptive profile is slightly but consistently better because stronger regularization reduces late-stage overfitting. Appendix Figures 1 and 2 therefore show the full retained configuration set: paper profile at 10/20/30 events and adaptive profile at 50/100 events.

These diagnostics justify the profile split. The short-horizon paper-profile losses remain well behaved across 10, 20, and 30 events, which argues against a high-bias explanation for the short-horizon errors. Instead, the main short-horizon failure mode is class imbalance: under the wrong checkpointing rule the model can drift toward the stationary class. By contrast, the adaptive profile is most useful at 50 and 100 events, where the retained runs show flatter late-stage validation behavior and cleaner diagonals. This is more consistent with a high-variance problem: the paper-style long-horizon runs fit the training signal but generalize slightly worse, while the adaptive profile reduces that variance through stronger regularization and stricter monitoring.

The baseline and factor analyses make the contribution more credible. Out of 104 engineered features, 21 survive the qualification pipeline and 20 of them belong to the price-and-volume derivative family. Ridge logistic regression reaches only accuracy 0.710 with near-zero κ, while a two-layer MLP improves to MCC 0.191. DeepLOB reaches accuracy 0.8347 and MCC 0.6080 at the same 10-event horizon, showing that the full temporal architecture extracts structure beyond what a static feature snapshot can recover.

5.2 Optiver Transfer Results

For the Optiver task, we report only the retained rolling-quantile mode in the main text because it produced the clearest and most stable transfer behavior. Table 2 summarizes the results across held-out stocks 42, 17, and 82.

Table 2: Optiver 5-event horizon results under rolling-quantile labels, averaged across held-out stocks 42, 17, and 82.

Regime	Mean accuracy	Mean κ	Mean MCC
Universal zero-shot	0.4211	0.0618	0.0626
Per-stock fine-tuned	0.4110	**0.0844**	**0.0847**
Specific-stock OOS	0.3700	0.0021	0.0021

The first rigorous conclusion is that **accuracy is not the right standalone summary for Optiver**. Fine-tuning slightly decreases mean accuracy from 0.4211 to 0.4110, yet improves both κ

and MCC materially. That means the fine-tuned model is matching the class distribution more faithfully and achieving better agreement beyond chance, even though it is no longer optimizing the most frequent class as aggressively. Per-stock values are reported in Appendix Table 3.

The retained rolling3 diagnostics support the same interpretation. The representative transfer loss curve in Appendix Figure 3 shows that validation macro-F1 peaks early and then oscillates within a narrow range, which justifies monitor-based checkpointing rather than simply training longer. This pattern does not look like classical high bias: the model can fit the task well enough for validation performance to peak quickly. Instead, the limitation is generalization under distribution shift. The universal base confusion matrix in Appendix Figure 4 shows that the model predicts all three classes but still over-predicts the stationary label. Per-stock fine-tuning mainly acts as a calibration step that reduces this mismatch.

Accuracy is especially fragile here for three reasons. First, it depends strongly on the instantaneous class mix, which changes across stocks and time. Second, the rolling-quantile labels are designed to stabilize return-scale differences rather than to maximize raw accuracy. Third, a conservative classifier can preserve accuracy by leaning toward the stationary class even when it gives poor directional agreement. For those reasons, we treat κ and MCC as primary evaluation metrics on Optiver and treat accuracy as supplementary.

The cumulative normalized profit plot in Appendix Figure 5 provides a third type of evidence. It shows six trajectories: zero-shot and transfer performance on each of the three held-out stocks. The curves are positive for both regimes on all three stocks, but the relative ranking is mixed. Transfer finishes above zero-shot on stocks 42 and 82, while stock 17 shows a lower ending curve despite a stronger κ after fine-tuning. This is exactly why no single scalar metric should dominate the interpretation. The profit proxy is not a tradable PnL claim and does not include transaction costs, but it is still useful as a directional diagnostic on held-out windows.

The local explanation in Appendix Figure 6 adds a final qualitative check. It focuses on a representative stationary prediction from stock 82 at horizon $k = 5$, showing the input window together with local importance heatmaps for the base and fine-tuned models. Because the saved artifact is a local-importance heatmap rather than a classical sparse LIME bar chart, we interpret it as qualitative evidence rather than definitive attribution. Even so, the post-transfer importance pattern is smoother and less dominated by a single block of the input, which is consistent with the view that fine-tuning mainly recalibrates the existing representation rather than relearning the signal from scratch.

6 Safety, Security, and Ethics

If a model like this were deployed at scale, the key risk would not be benchmark accuracy but failure under regime shift. A model that appears predictive in one period may become unstable in another, especially in markets with changing liquidity, event intensity, or volatility. In this project, the near-zero specific-stock out-of-sample results are a direct warning sign: the model can appear calibrated on one segment and fail badly on the next.

Security concerns are also concrete. Financial data pipelines are vulnerable to preprocessing mistakes, temporal leakage, silent distribution drift, and adversarial behavior such as spoofing-like order placement that changes the local book without reflecting durable supply or demand. Because our Optiver workflow relies on causal normalization and rolling-quantile labels, even a small violation of temporal ordering could make the results look much stronger than they truly are. Reproducibility, audit trails, and strict chronological validation are therefore part of the security story, not just the engineering story.

There are also fairness and misuse concerns. Access to low-latency data and large-scale compute is uneven, so advanced market models can reinforce structural inequalities in who benefits from automation. If a model like this were presented as decision support without uncertainty communication, less sophisticated users could be exposed to disproportionate risk. For that reason, the present results should be interpreted as a research study of representation learning under shift, not as evidence that the model is ready for live deployment.

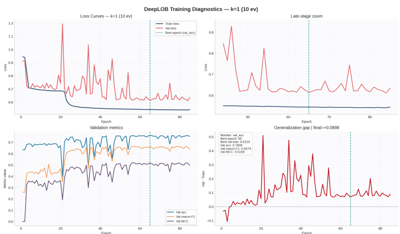
Reasonable mitigations would include abstention near the decision boundary, explicit drift monitoring, conservative deployment thresholds, and human oversight. Given the current Optiver transfer results, the correct scientific conclusion is caution: the model is informative, but not deployment-ready.

7 Conclusion

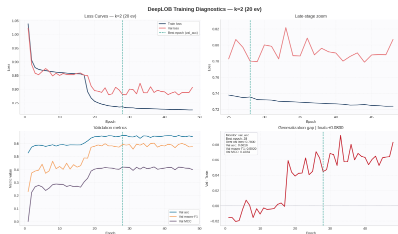
This project shows that DeepLOB can be reproduced credibly on FI-2010 and partially transferred to a different market representation, but the reasons for success differ across datasets. On FI-2010, the architecture itself is strong and the main lesson is that training control matters: the paper-style profile dominates at short horizons, while the adaptive profile is slightly better at long horizons. On Optiver, the architecture still captures useful signal, but the decisive improvements come from label design and calibration through per-stock fine-tuning.

The broader lesson is methodological. For real financial sequence data, model architecture is only one part of performance. Label construction, evaluation metrics, and temporal validation protocol matter just as much. The main contribution of the project is therefore not a claim of strong trading accuracy, but a clearer account of what transfers, what does not, and how to evaluate a deep model responsibly under nonstationarity.

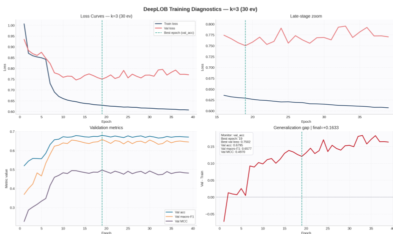
A Appendix: FI-2010 Diagnostic Figures



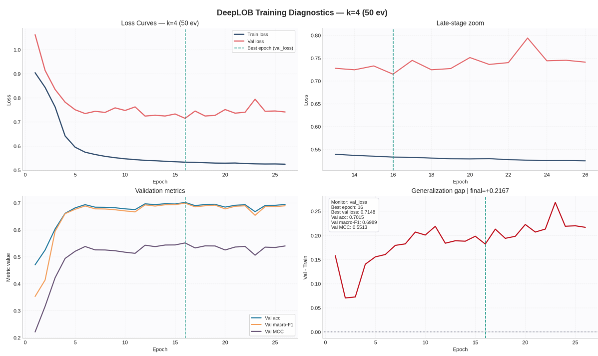
(a) 10 events, paper



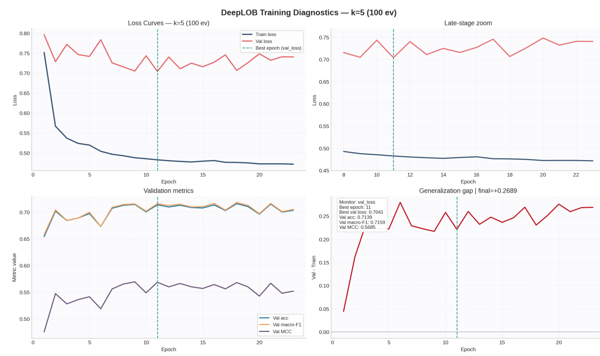
(b) 20 events, paper



(c) 30 events, paper

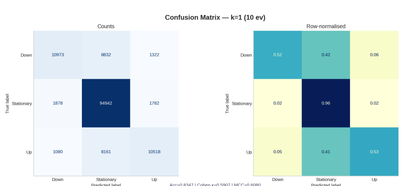


(d) 50 events, adaptive

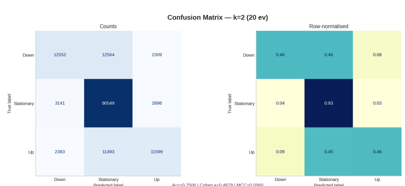


(e) 100 events, adaptive

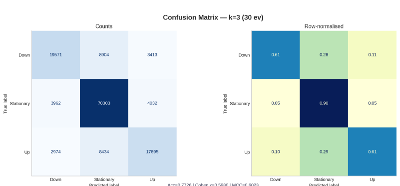
Figure 1: FI-2010 loss curves for the retained profile at each reported horizon: paper profile for 10/20/30 events and adaptive profile for 50/100 events. Together these curves document the exact configuration split used in Table 1.



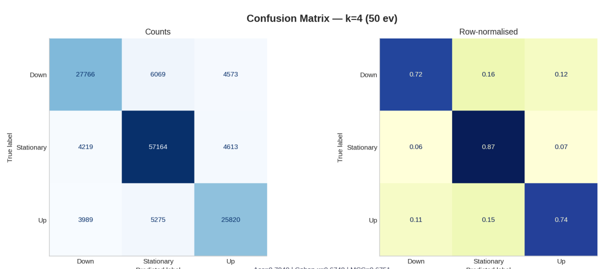
(a) 10 events, paper



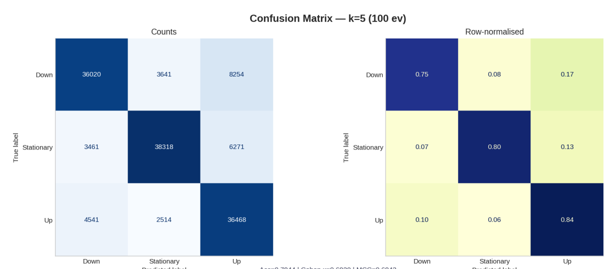
(b) 20 events, paper



(c) 30 events, paper



(d) 50 events, adaptive



(e) 100 events, adaptive

Figure 2: FI-2010 confusion matrices for the retained profile at each reported horizon. The short-horizon paper runs preserve directional recall, while the long-horizon adaptive runs show the cleanest diagonals.

B Appendix: Optiver Diagnostic Figures

Table 3: Per-stock Optiver results at the 5-event horizon under rolling-quantile labels.

Stock	Zero-shot		Fine-tuned		Specific OOS
	Acc.	κ	Acc.	κ	κ
42	0.4214	0.0697	0.4044	0.0752	-0.0048
17	0.4181	0.0388	0.4033	0.0841	0.0066
82	0.4239	0.0769	0.4252	0.0938	0.0045

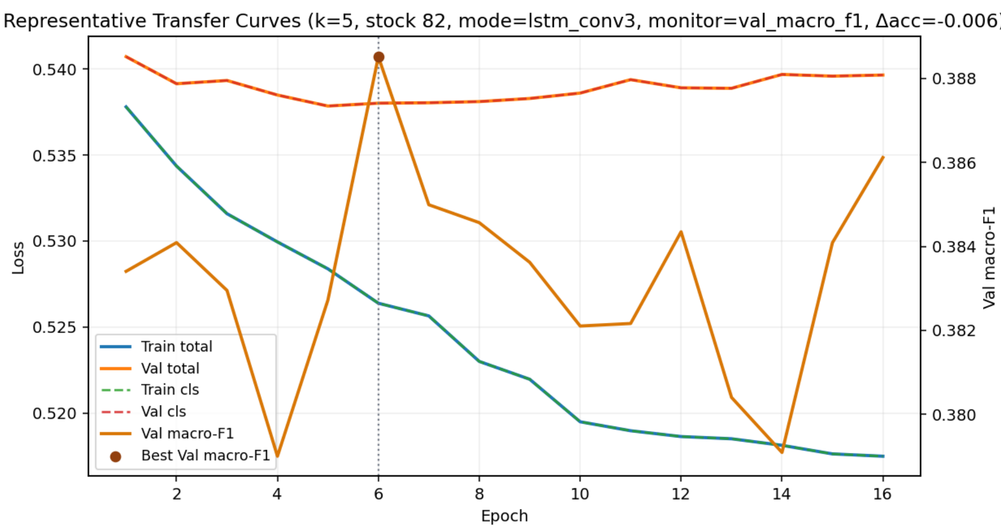


Figure 3: Representative Optiver transfer loss curve under the retained rolling3 setup. Validation macro-F1 peaks early and then stabilizes, which supports checkpointing by validation macro-F1 rather than training for more epochs.

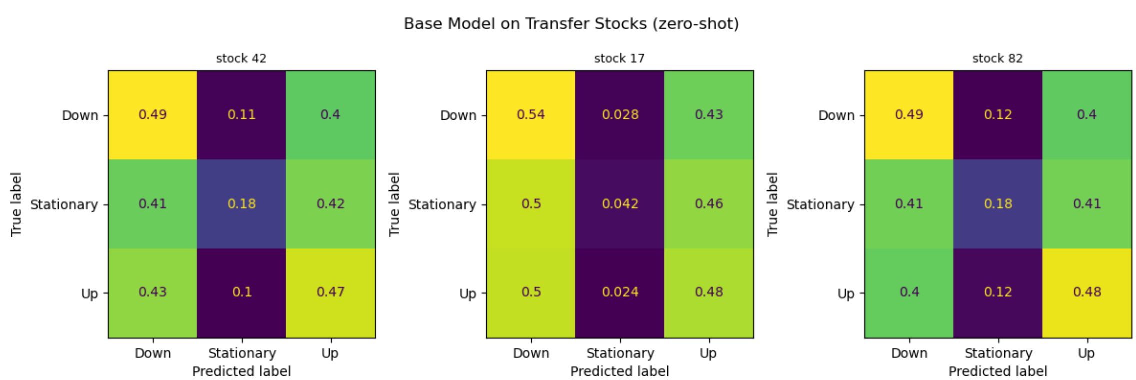


Figure 4: Optiver universal base-model confusion matrix on held-out stocks. The model predicts all classes but still overweights the stationary class, motivating the use of κ and MCC in addition to accuracy.

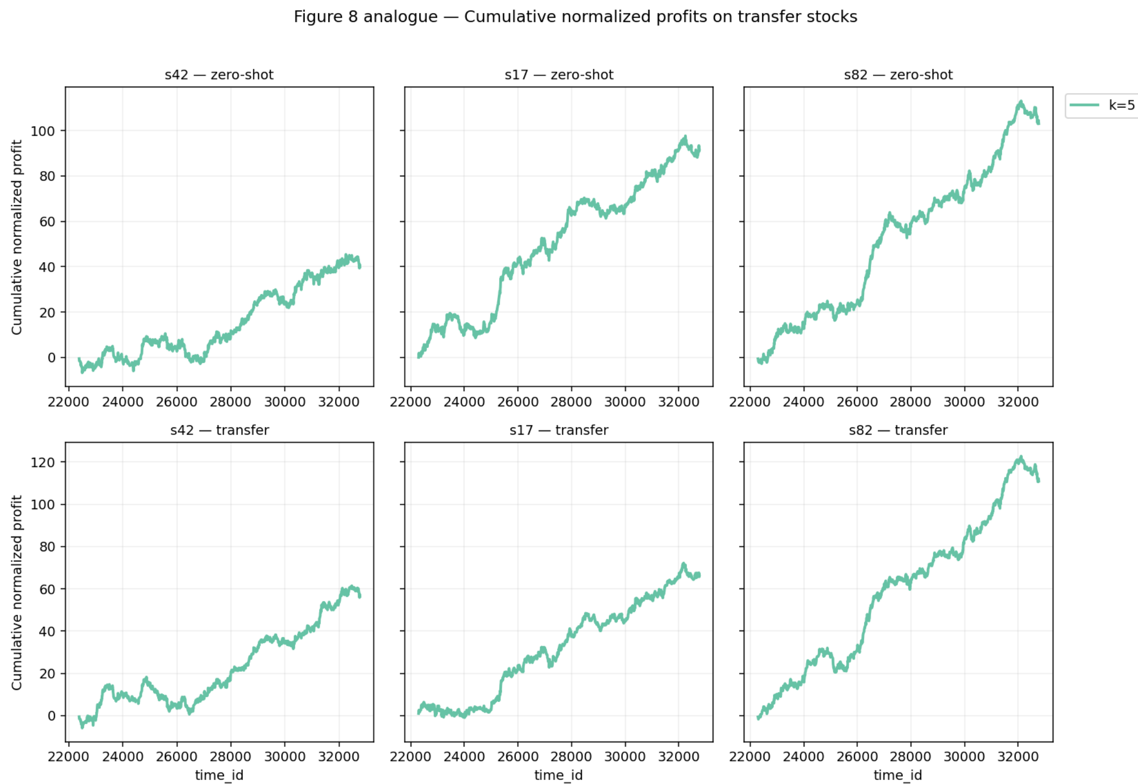


Figure 5: Cumulative normalized profit trajectories for the three held-out stocks under zero-shot transfer and per-stock fine-tuning. These curves are diagnostic only and do not represent executable, transaction-cost-adjusted trading PnL.

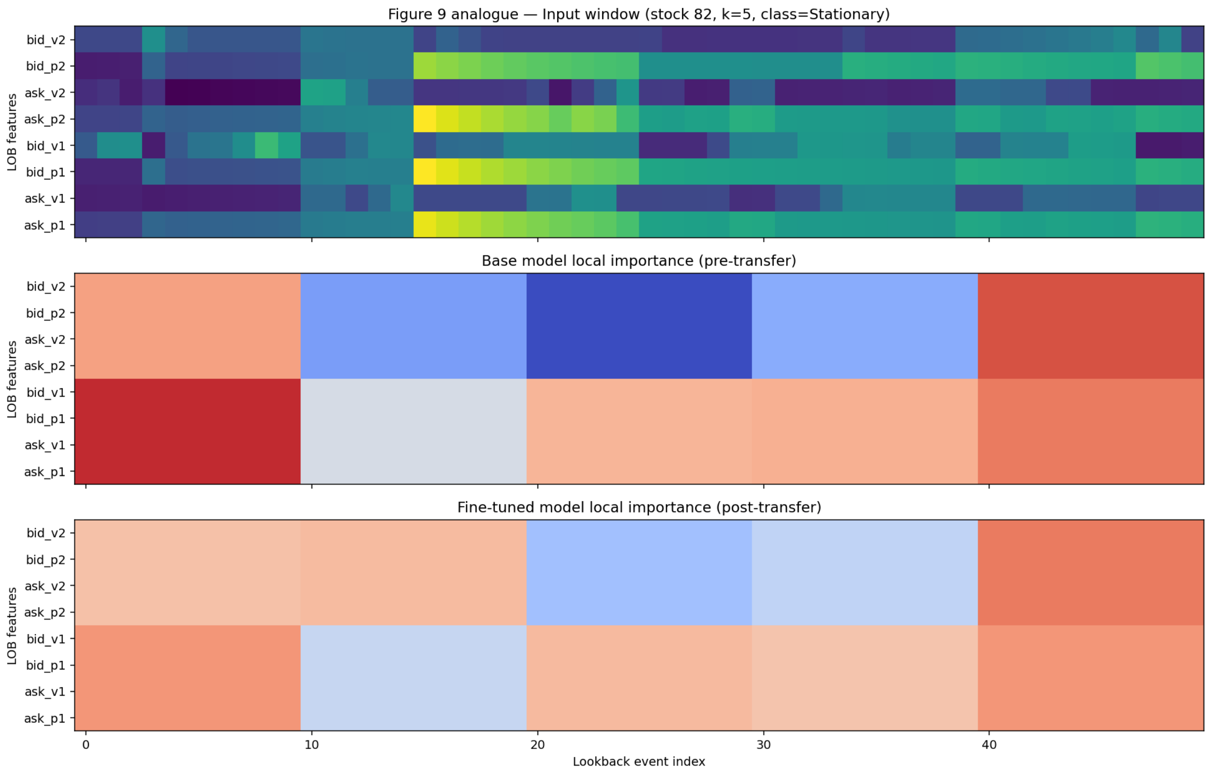


Figure 6: LIME-style local explanation for a representative stationary Optiver sample from stock 82 at horizon $k = 5$. The top panel shows the input window; the lower panels compare the local importance structure before and after transfer fine-tuning.

C Appendix: Code Availability

Code, training scripts, and the cleaned report artifacts are available in the public GitHub repository:

https://github.com/xiao-113/IEOR242-26Spring-finalproject

The FI workflow is primarily driven by `scripts/train_deeplob.py`, while the Optiver workflow is driven by `scripts/train_optiver.py`. We provide the full code in the repository rather than reproducing long code listings in the report.

References

[1] Alec N. Kercheval and Yuan Zhang. Modelling high-frequency limit order book dynamics with support vector machines. *Quantitative Finance*, 15(8):1315–1329, 2015.

[2] Apostolos Ntakaris, Mikko Magris, Juho Kanniainen, Moncef Gabbouj, and Alexandros Iosifidis. Benchmark dataset for mid-price forecasting of limit order book data. *Journal of Forecasting*, 37(8):852–866, 2018.

[3] Optiver and Kaggle. Optiver realized volatility prediction. https://www.kaggle.com/competitions/optiver-realized-volatility-prediction, 2021.

[4] Zihao Zhang, Stefan Zohren, and Stephen Roberts. Deeplob: Deep convolutional neural networks for limit order books. *IEEE Transactions on Signal Processing*, 67(11):3001–3012, 2019.